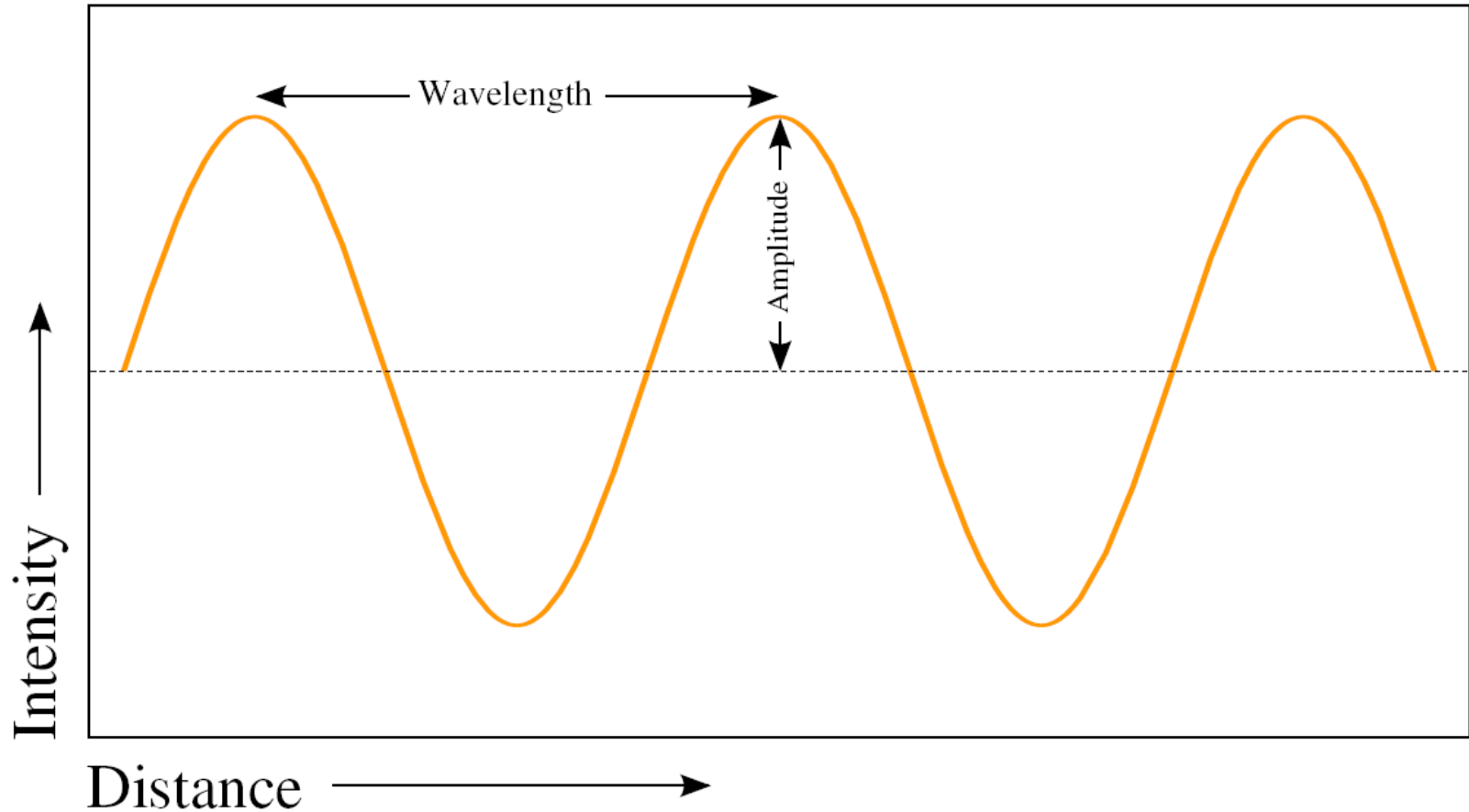


2 Digital Image Fundamentals

- Electromagnetic radiation (Electromagnetic wave) can be thought of as a stream of massless particles, called photons, that carry energy in straight-line paths through space.
- They travel in a wavelike (sinusoidal) pattern and move through empty space at the speed of light.
- They travel somewhat more slowly through air, water, glass, or other media.
- Each massless particle contains a certain amount (or bundle) of energy.

Electromagnetic Wave



Electromagnetic Energy

- Electromagnetic energy is proportional to frequency ($E=h\nu$)
- The higher frequency (shorter wavelength) electromagnetic phenomena carry more energy per photon.

Electromagnetic Spectrum

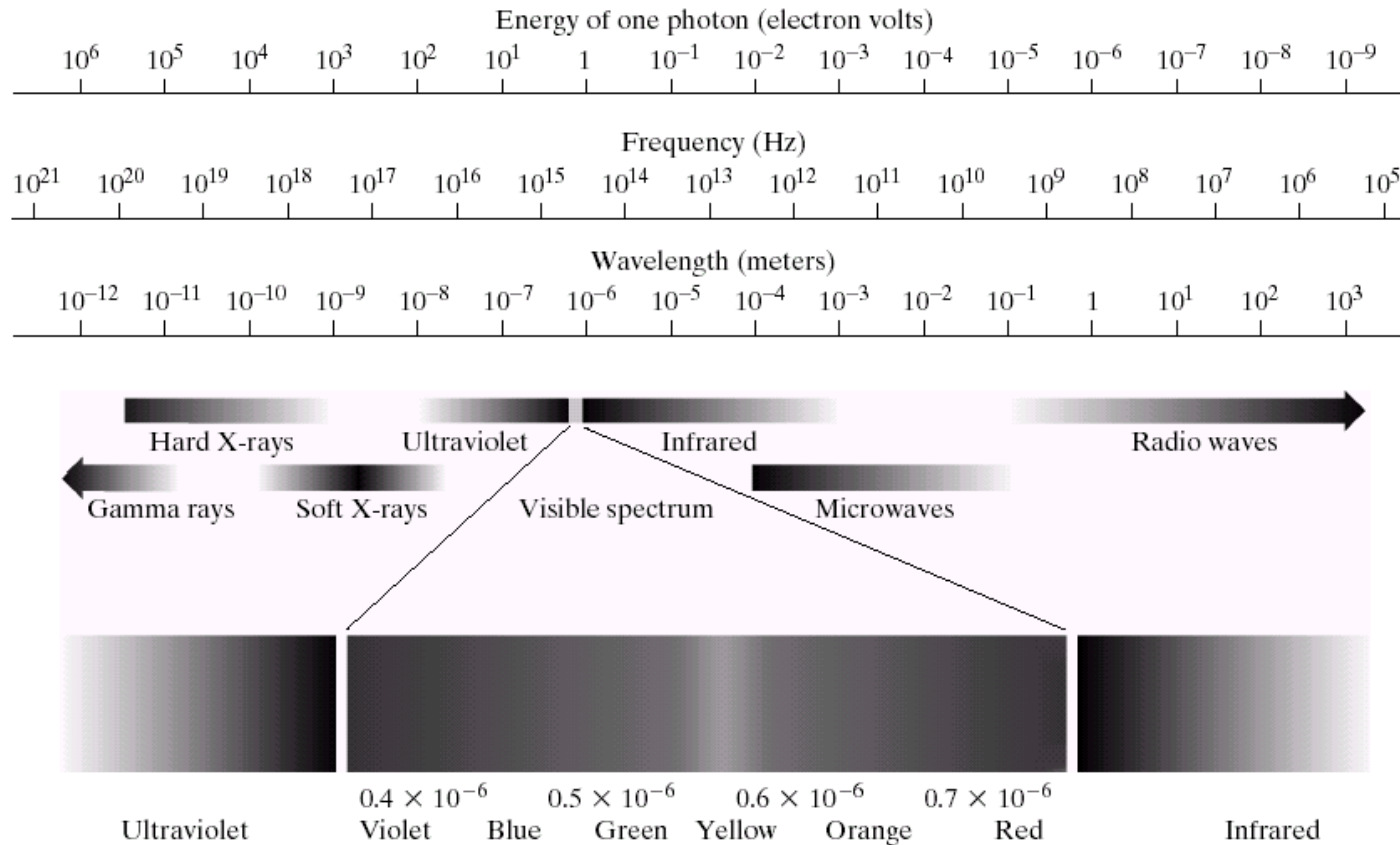


FIGURE 2.10 The electromagnetic spectrum. The visible spectrum is shown zoomed to facilitate explanation, but note that the visible spectrum is a rather narrow portion of the EM spectrum.

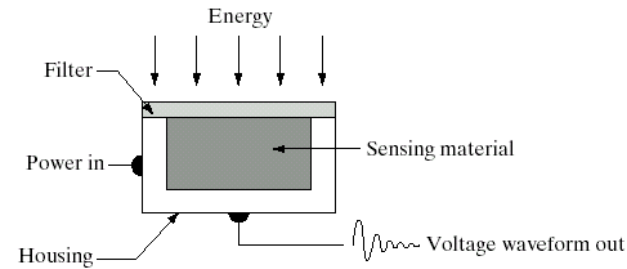
Light and Color Sensing

- Light is a particular type of electromagnetic radiation that can be seen and sensed by the human eye.
- Chromatic light spans the electromagnetic energy spectrum from approximately 0.43 to 0.79 μm .
- The colors that humans perceive in an object are determined by the nature of the light *reflected* from the object.

Quality of a Chromatic Light

- Three basic quantities are used to describe the quality of a chromatic light source:
 - Radiance
 - *Radiance* is the total amount of energy that flows from the light source, and it is usually measured in watts (W).
 - Luminance
 - *Luminance*, measured in lumens (lm), gives a measure of the amount of energy an observer *perceives* from a light source.
 - Brightness.
 - *Brightness* is a subjective descriptor of light perception that is practically impossible to measure.
 - It embodies the achromatic notion of intensity and is one of the key factors in describing color sensation.

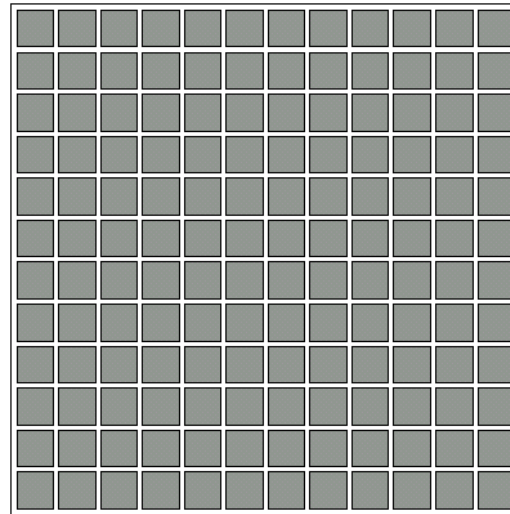
Image Sensing and Acquisition



Single Sensor



Line Sensor



Array Sensor

Image Sensing and Acquisition

- Incoming energy is transformed into a voltage by the combination of input electrical power and sensor material that is responsive to the particular type of energy being detected.
- The output voltage waveform is the response of the sensor(s), and a digital quantity is obtained from each sensor by digitizing its response.

Image Sensor

- The two main types of digital image sensors are the charge-coupled device (CCD) and the active-pixel sensor (CMOS sensor).
- Both CCD and CMOS sensors are based on MOS technology.
- MOS capacitors being the building blocks of a CCD.
- MOSFET amplifiers being the building blocks of a CMOS sensor.

Image Acquisition Using a Single Sensor

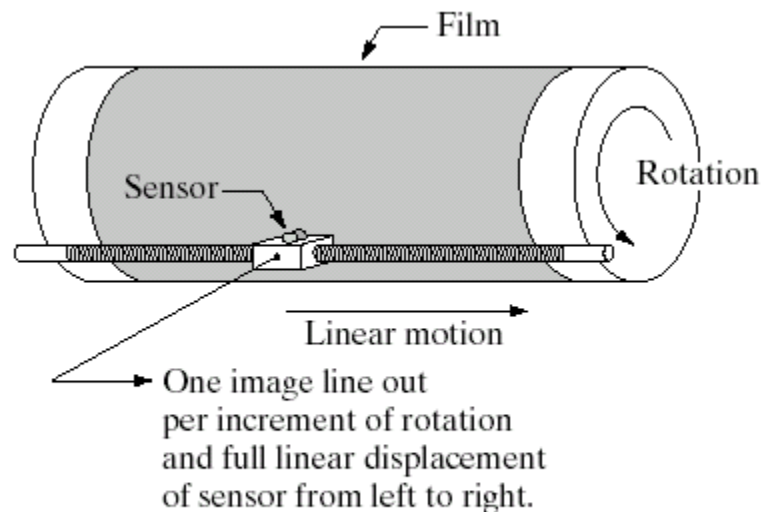
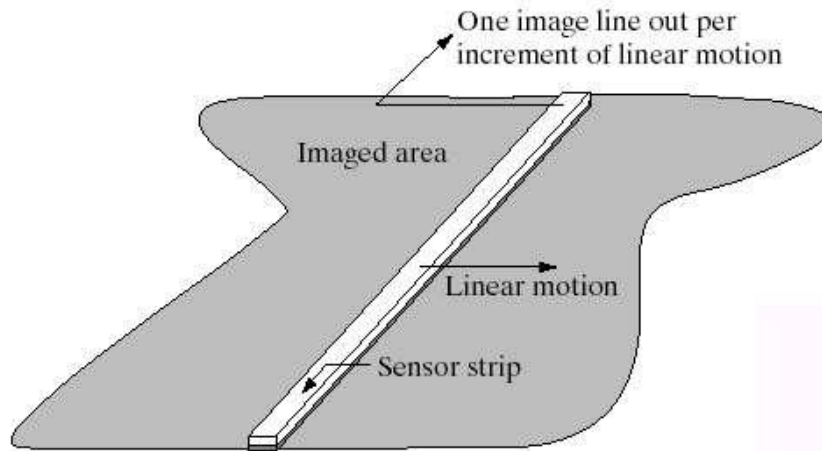
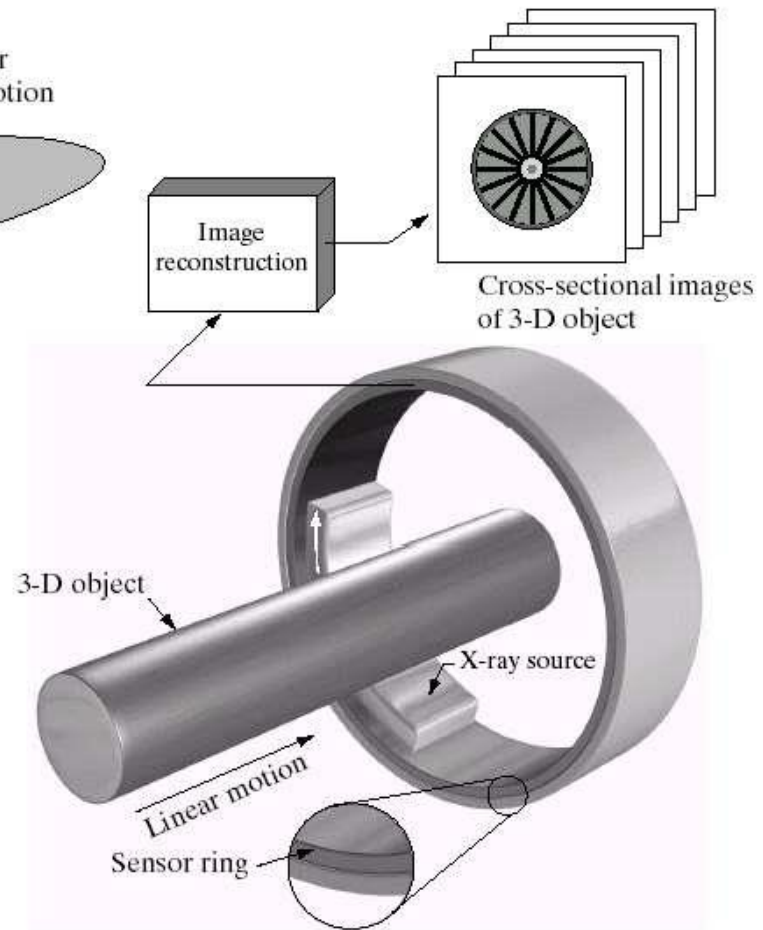


FIGURE 2.13 Combining a single sensor with motion to generate a 2-D image.

Image Acquisition Using Sensor Strips



Fingerprint sweep sensor



Computerized Axial Tomography (CAT)

Image Acquisition Using Sensor Arrays

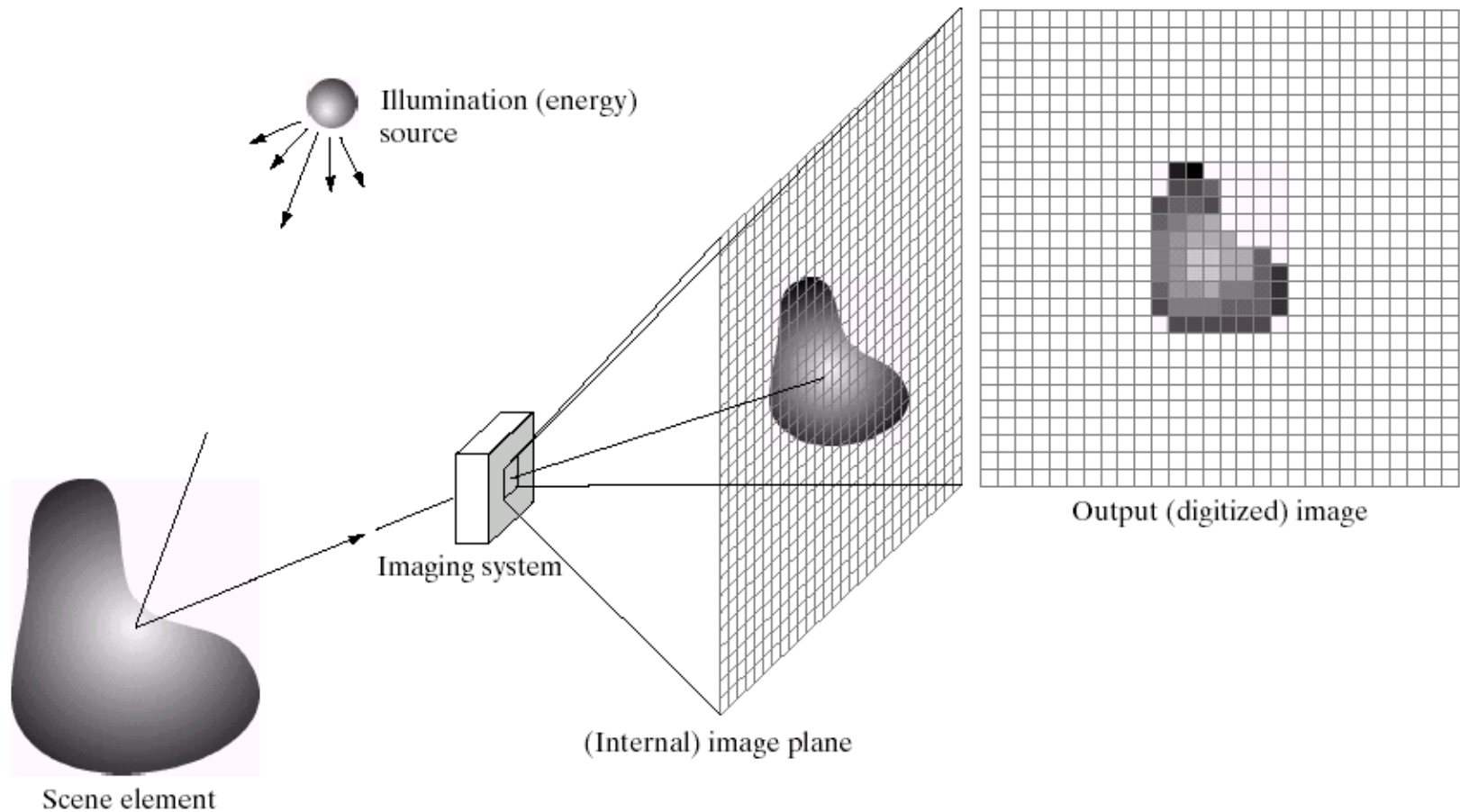


Image Acquisition Using Digital Camera

- The sunlight is used as a source of energy.
- When the sunlight falls upon the object, then the object reflects light.
- The front end of the imaging system is a lens, which projects the viewed scene onto the lens focal plane.
- The sensor array is coincident with the focal plane.
- The sensor array produces outputs (analog voltage signal) proportional to the integral of the light received at each sensor.
- This data from each sensor is converted into a digital form. This involves sampling and quantization.

Energy Radiation

- The incoming energy is characterized by two components:
 - The amount of source illumination incident on the scene being viewed, denoted by i .
 - The amount of illumination reflected by the objects in the scene, denoted by r .
- The incoming energy is $i*r$.

Image Sampling and Quantization

- The output of most sensors is a continuous voltage waveform whose amplitude and spatial behavior are related to the physical phenomenon being sensed.
- Digitizing the coordinate values is called *sampling*.
- Digitizing the amplitude values is called *quantization*.

Image Sampling and Quantization

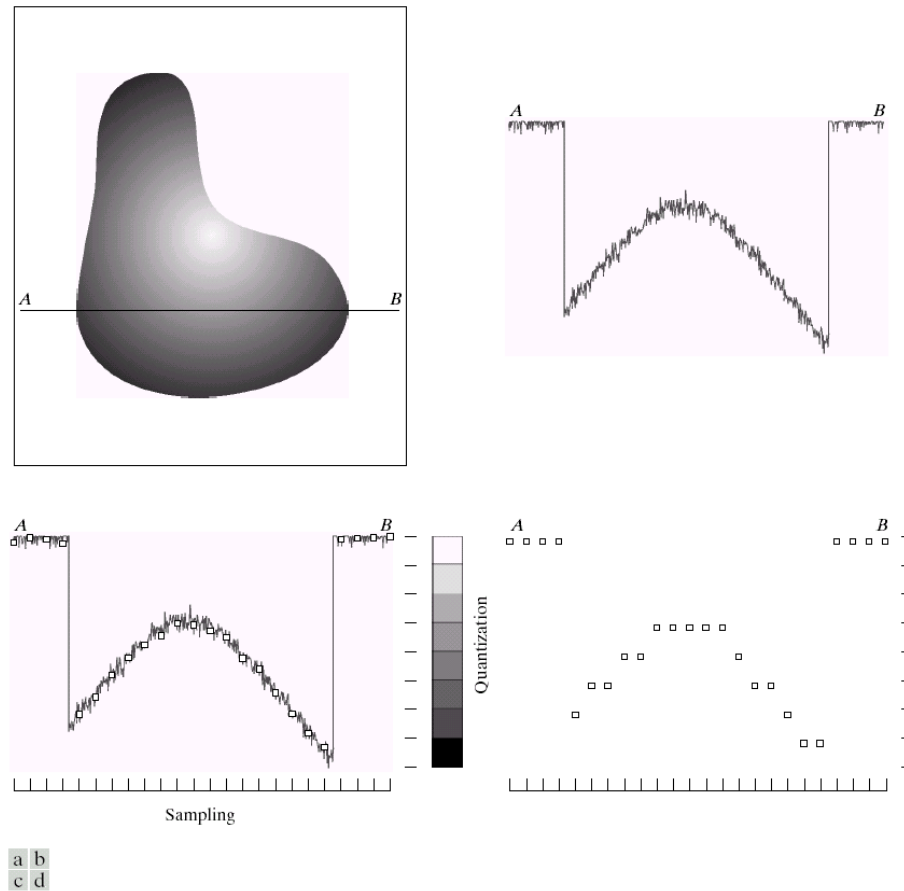
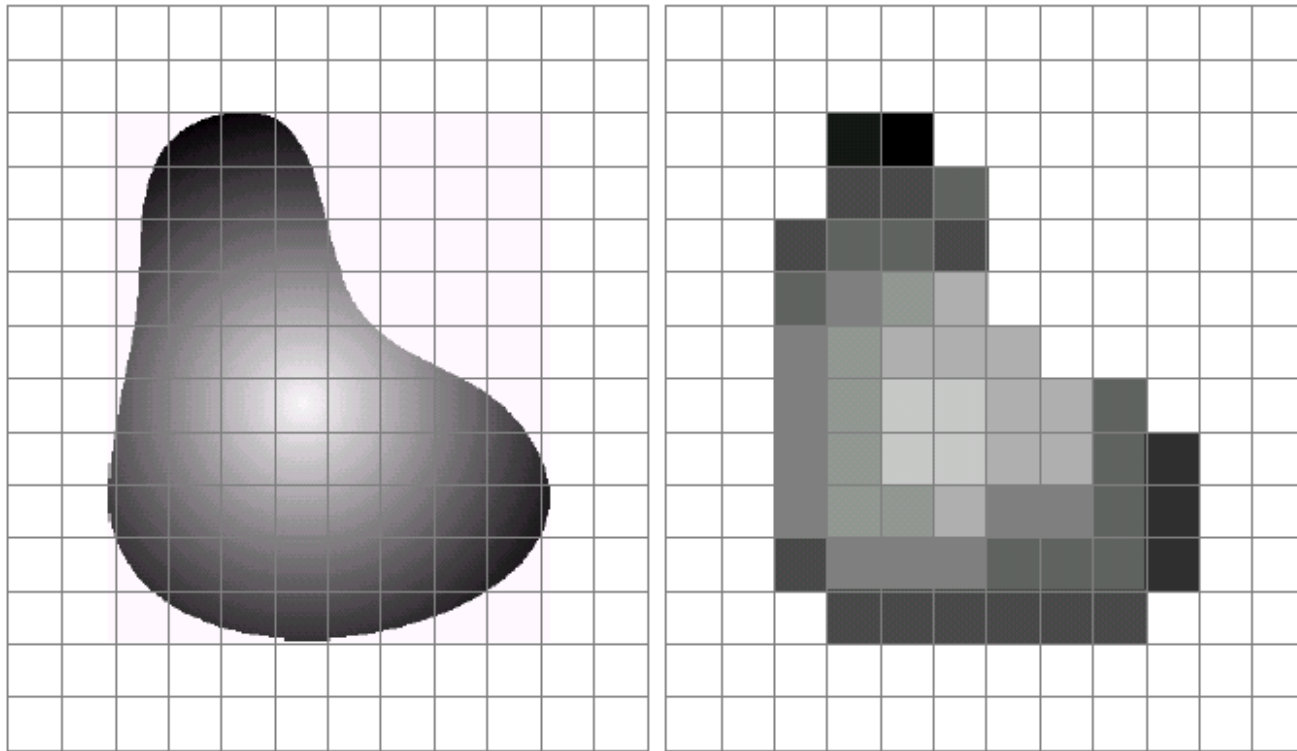


FIGURE 2.16 Generating a digital image. (a) Continuous image. (b) A scan line from *A* to *B* in the continuous image, used to illustrate the concepts of sampling and quantization. (c) Sampling and quantization. (d) Digital scan line.

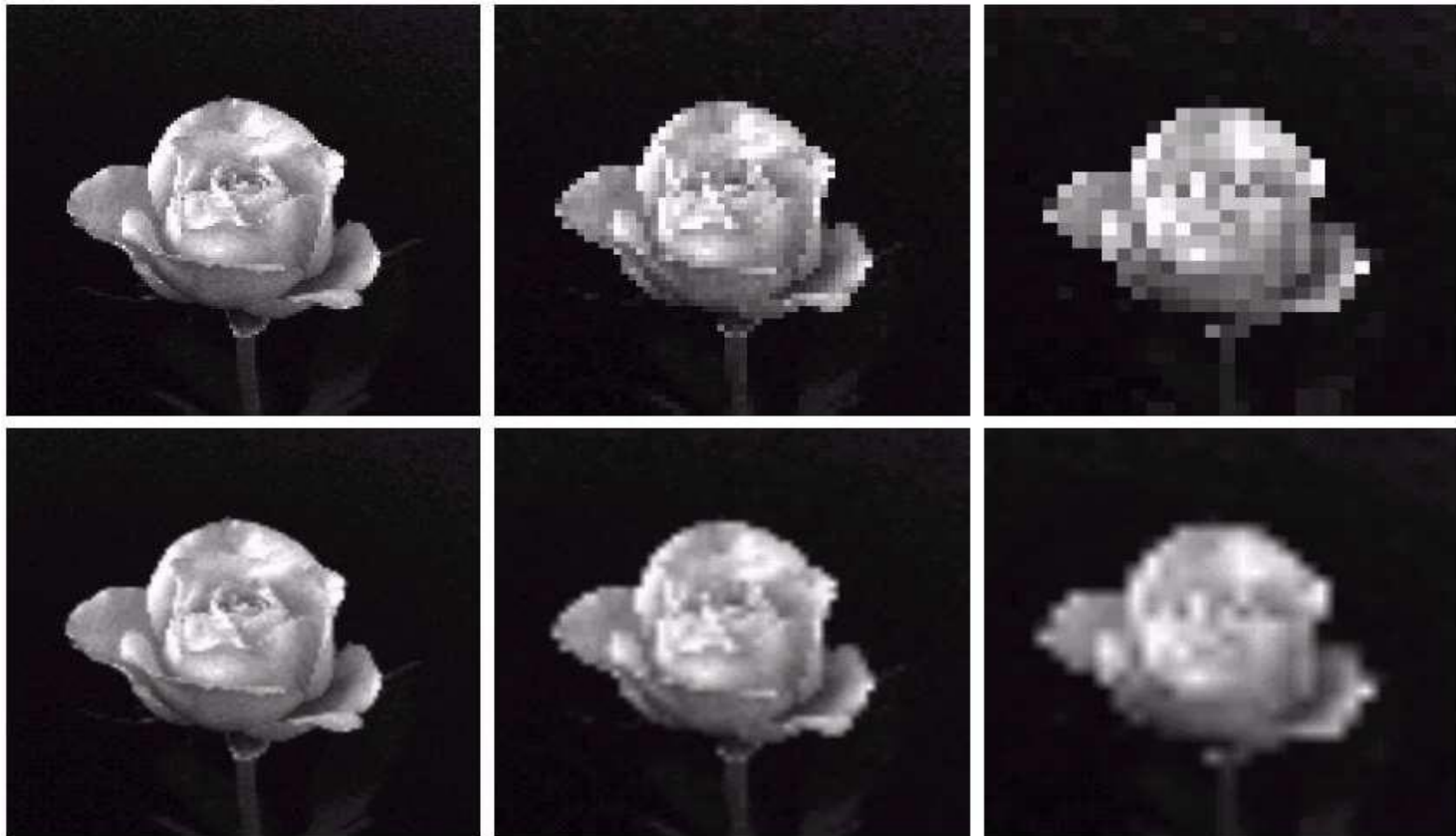
Image Sampling and Quantization



a b

FIGURE 2.17 (a) Continuous image projected onto a sensor array. (b) Result of image sampling and quantization.

Zooming



a	b	c
d	e	f

FIGURE 2.25 Top row: images zoomed from 128×128 , 64×64 , and 32×32 pixels to 1024×1024 pixels, using nearest neighbor gray-level interpolation. Bottom row: same sequence, but using bilinear interpolation.

Zooming

- Zooming means oversampling.
- Zooming requires two steps:
 - The creation of new pixel locations.
 - The assignment of gray levels to those new locations.
- Three zooming methods:
 - Nearest neighbor interpolation
 - Pixel replication
 - Bilinear interpolation

Zooming: Nearest Neighbor Interpolation

- Create new dimensional grid over the original image.
- The spacing in the new grid would be less than one pixel because we are fitting it over a smaller image.
- For any point in the overlay, look the closest pixel in the original image and assign its gray level to the new pixel in the grid.

Shrinking

- Shrinking means undersampling.
- The shrinking is accomplished by deleting the appropriate number of rows and columns from the original image.

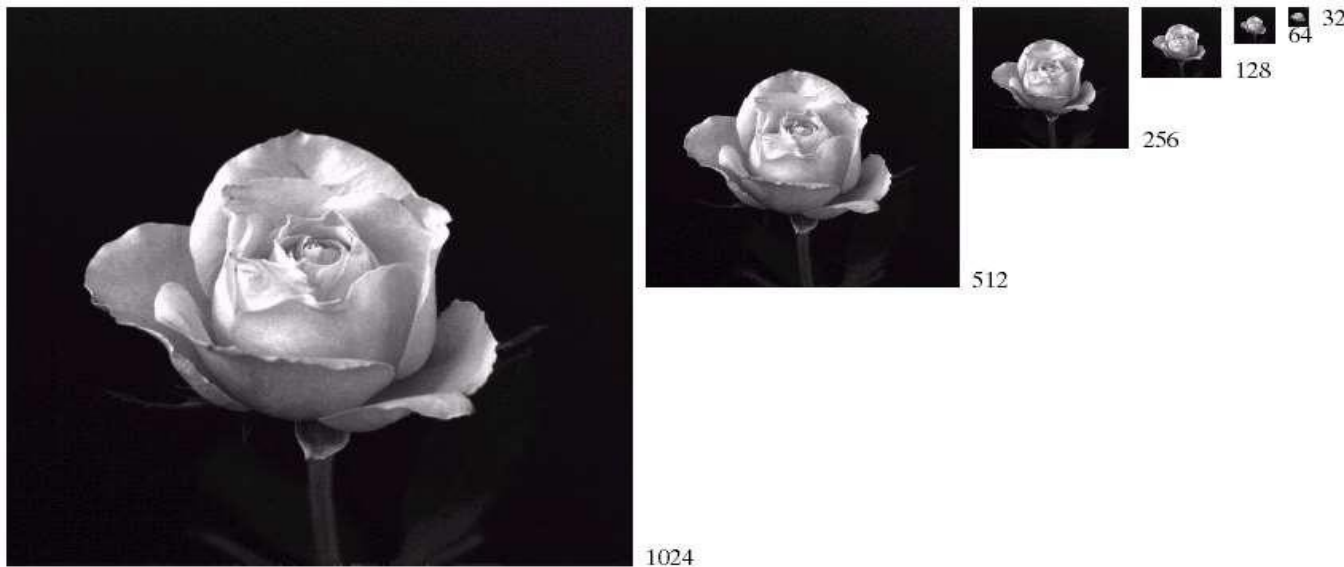
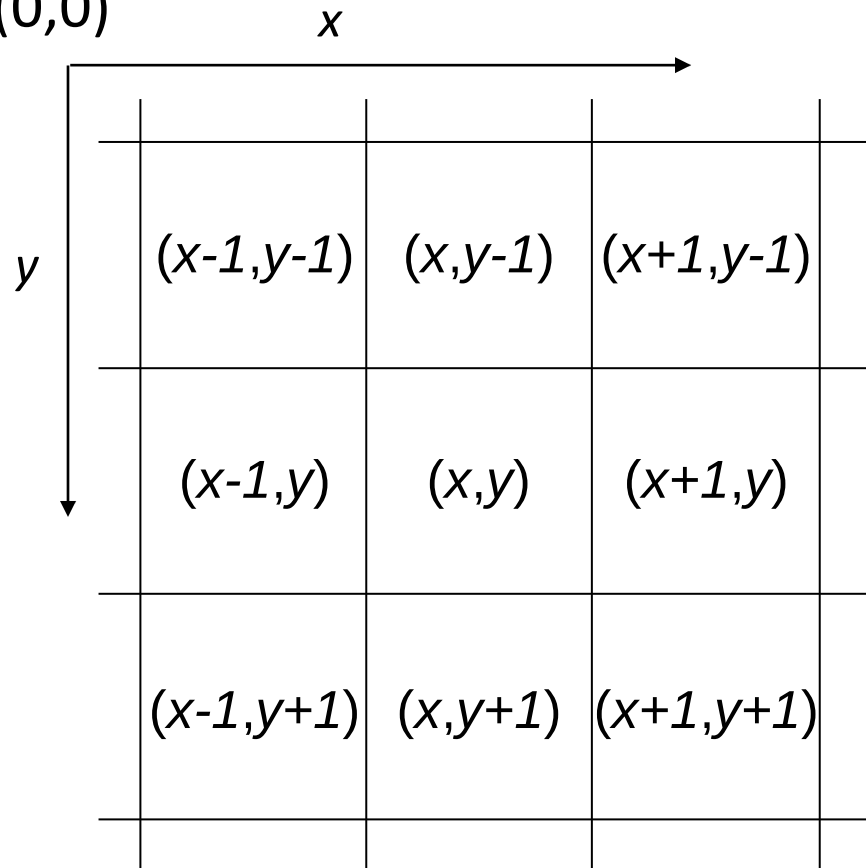


FIGURE 2.19 A 1024×1024 , 8-bit image subsampled down to size 32×32 pixels. The number of allowable gray levels was kept at 256.

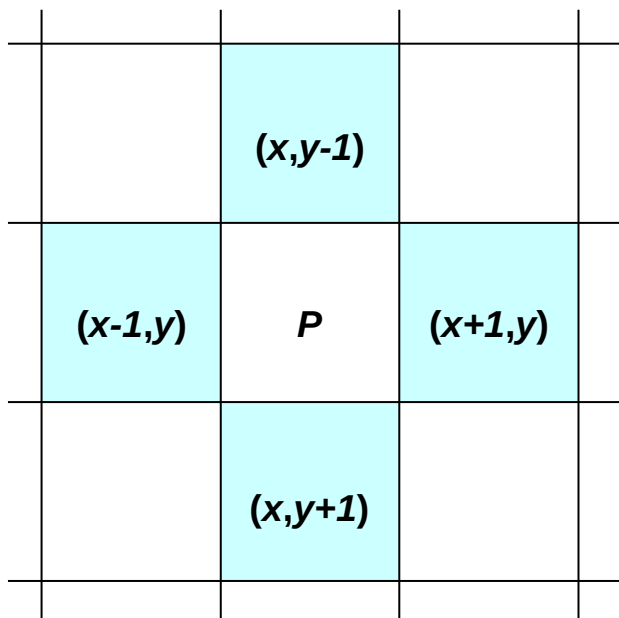
Relationships Between Pixels

- $(0,0)$



Neighbors of a Pixel

- Neighborhood relation is used to determine adjacent pixels. It is useful for analyzing regions.



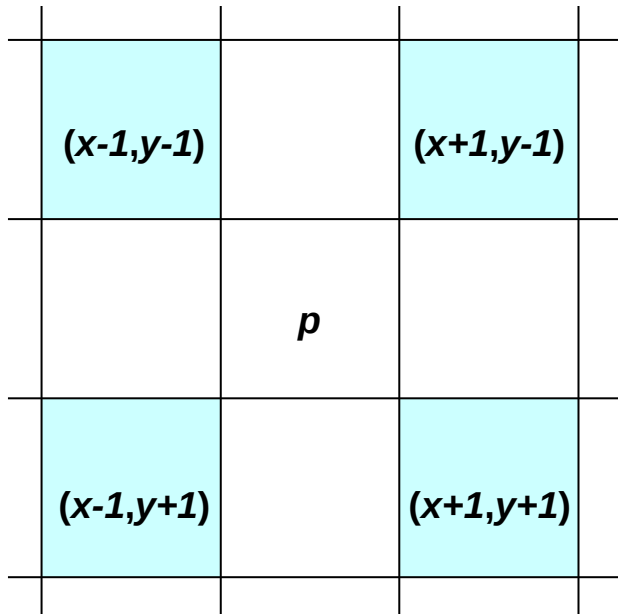
4-neighbors of p :

$$N_4(p) = \left\{ \begin{array}{l} (x-1,y) \\ (x+1,y) \\ (x,y-1) \\ (x,y+1) \end{array} \right\}$$

4-neighborhood relation considers only vertical and horizontal neighbors.

Note: $q \in N_4(p)$ implies $p \in N_4(q)$

Neighbors of a Pixel



Diagonal neighbors of p :

$$N_D(p) = \left\{ \begin{array}{l} (x-1, y-1) \\ (x+1, y-1) \\ (x-1, y+1) \\ (x+1, y+1) \end{array} \right\}$$

Diagonal -neighborhood relation considers only diagonal neighbor pixels.

Neighbors of a Pixel

$(x-1, y-1)$	$(x, y-1)$	$(x+1, y-1)$
$(x-1, y)$	p	$(x+1, y)$
$(x-1, y+1)$	$(x, y+1)$	$(x+1, y+1)$

8-neighbors of p :

$$N_8(p) = \left\{ \begin{array}{l} (x-1, y-1) \\ (x, y-1) \\ (x+1, y-1) \\ (x-1, y) \\ (x+1, y) \\ (x-1, y+1) \\ (x, y+1) \\ (x+1, y+1) \end{array} \right\}$$

8-neighborhood relation considers all neighbor pixels,
i.e., $N_8(p) = N_4(p) \cup N_D(p)$

Adjacency and Connectivity

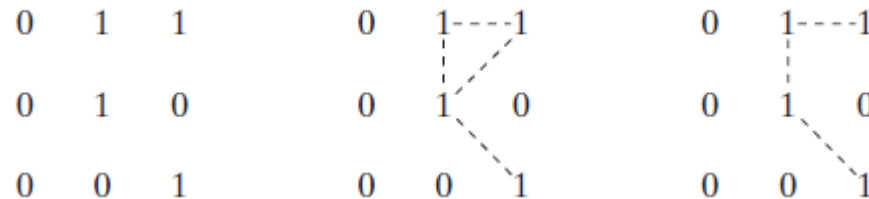
- Connectivity between pixels determines regions and boundaries.
- Two pixels are connected if they are neighbors and if their gray levels satisfy a specified criterion of similarity (say, if their gray levels are equal).

Adjacencies

1. **4-adjacency:** Two pixels p and q with values from V are 4-adjacent if q is in the set $N_4(p)$.
2. **8-adjacency:** Two pixels p and q with values from V are 8-adjacent if q is in the set $N_8(p)$.
3. **m -adjacency (mixed-adjacency):**
Two pixels p and q with values from V are m -adjacent if :
 - i. q is in $N_4(p)$, or
 - ii. q is in $N_D(p)$ and the set $N_4(p) \cap N_4(q)$ has no pixel whose values are from V (i.e., $N_4(p) \cap N_4(q) = \phi$) .

m-Adjacency

- *m*-adjacency is a modification of 8-adjacency.
- It is introduced to eliminate the ambiguities that often arise when 8-adjacency is used.
- For example:



a b c

FIGURE 2.26 (a) Arrangement of pixels; (b) pixels that are 8-adjacent (shown dashed) to the center pixel; (c) *m*-adjacency.

Distance Measures

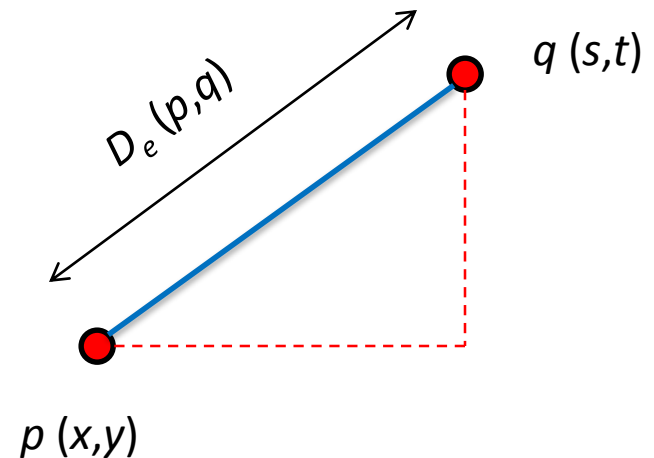
- For pixels p , q and z , with coordinates (x, y) , (s, t) and (v, w) , respectively, D is a distance function if:
 - (a) $D(p, q) \geq 0$ ($D(p, q) = 0$ iff $p = q$),
 - (b) $D(p, q) = D(q, p)$, and
 - (c) $D(p, z) \leq D(p, q) + D(q, z)$.

Euclidean Distance

- The *Euclidean Distance* between p and q is defined as:

$$D_e(p, q) = [(x - s)^2 + (y - t)^2]^{1/2}$$

- Pixels having a distance less than or equal to some value r from (x, y) are the points contained in a disk of radius r centered at (x, y) .

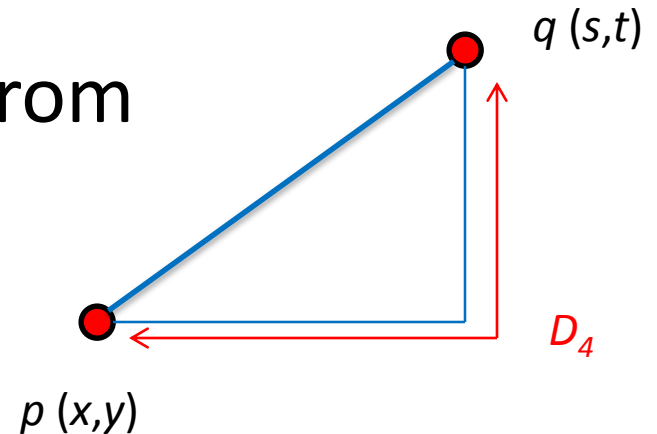


D_4 Distance Or City-Block Distance

- The D_4 distance (also called *city-block distance*) between p and q is defined as:

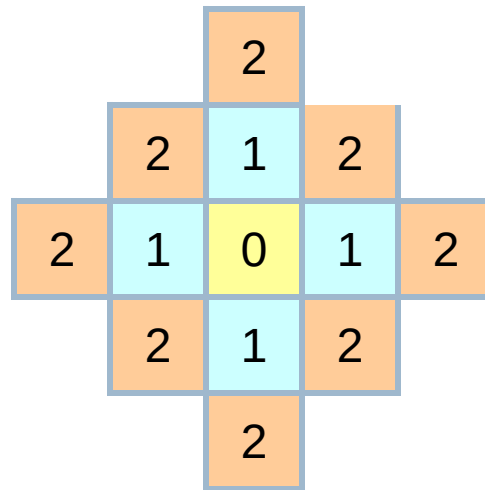
$$D_4(p, q) = |x - s| + |y - t|$$

- Pixels having a D_4 distance from (x, y) , less than or equal to some value r form a diamond centered at (x, y) .



D_4 Distance Or City-Block Distance

- The pixels with distance $D_4 \leq 2$ from (x, y) form the following contours of constant distance.
- The pixels with $D_4 = 1$ are the 4-neighbors of (x, y) .

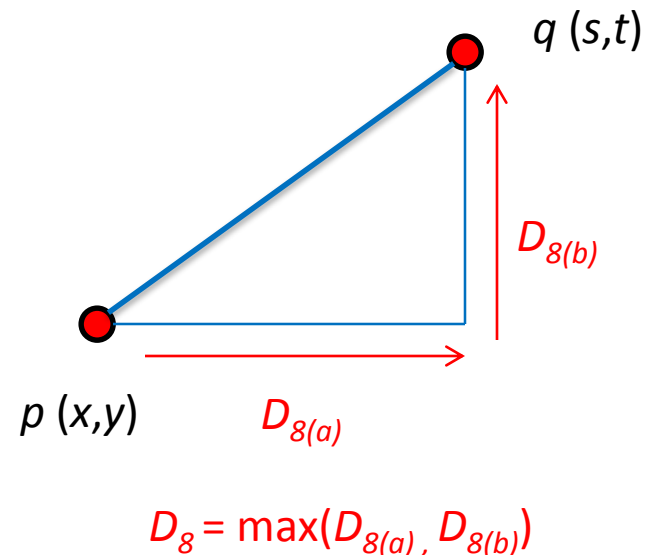


D_8 Distance Or Chessboard Distance

- The D_8 distance (also called *chessboard distance*) between p and q is defined as:

$$D_8(p, q) = \max(|x - s|, |y - t|)$$

- Pixels having a D_8 distance from (x, y) , less than or equal to some value r form a square centered at (x, y) .



D_8 Distance Or Chessboard Distance

- D_8 distance ≤ 2 from (x, y) form the following contours of constant distance.

2	2	2	2	2
2	1	1	1	2
2	1	0	1	2
2	1	1	1	2
2	2	2	2	2